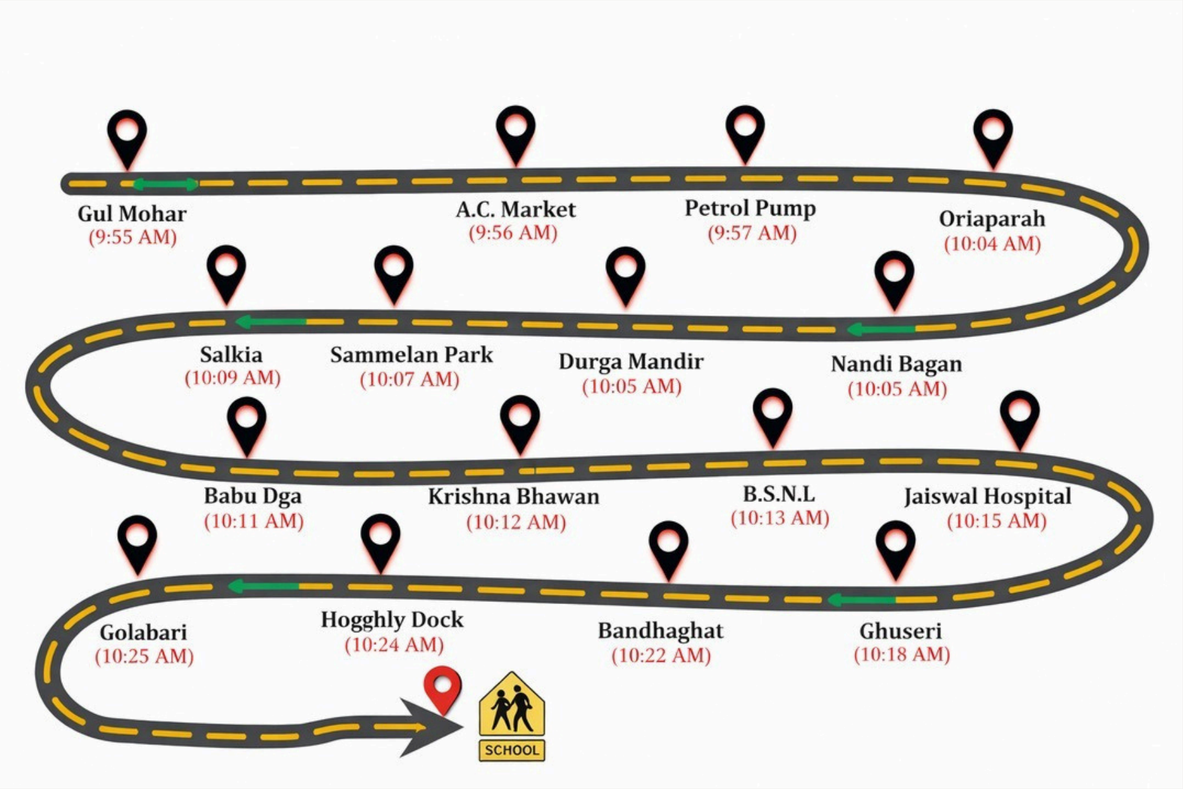
Probabilistic School Bus Routing with Stop Selection

Ahmedabad University



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Deterministic Baseline Overview

Baseline (Milestone 3)

Two deterministic heuristics implemented:

Nearest Neighbour Heuristic

Start at depot. At each step select the closest unvisited stop. Repeat until all stops visited. Return to depot.

Clarke-Wright Savings

Assign each stop its own route. Compute savings $S_{ij} = d_{i0} + d_{0j} - d_{ij}$. Merge routes in descending order of savings.

Key Limitations

1. **No Flexibility**

   Identical output for identical inputs — cannot adapt to real-world variation

2. **Greedy Behavior**

   Locally optimal choices lead to globally suboptimal routes

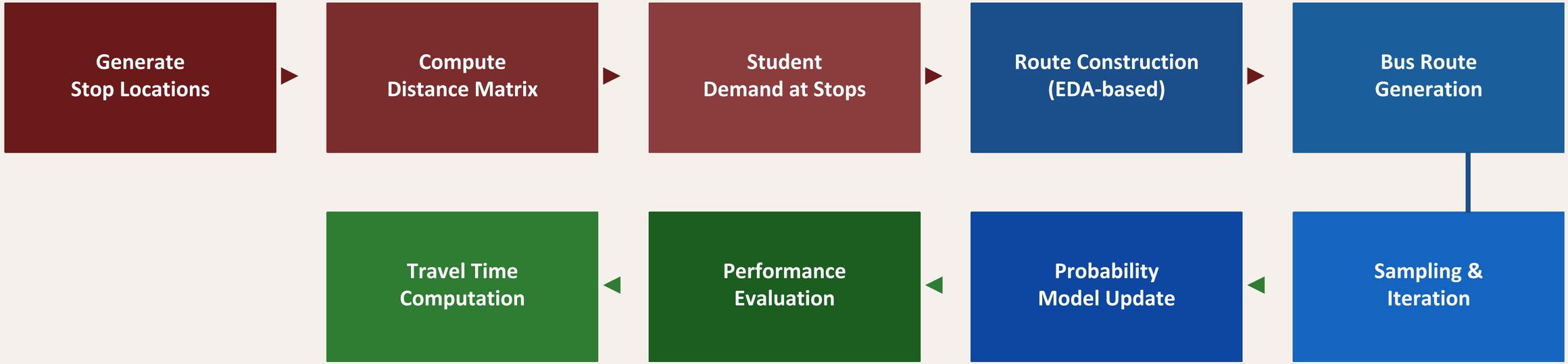
3. **Input Sensitivity**

   Small changes in stop locations or demand affect routes unpredictably

4. **Scalability Issues**

   Performance degrades as the number of stops and buses increases

System Pipeline Overview



Key Change in M4: Route Construction now uses EDA-based probabilistic sampling — replaces fixed greedy decision rules.

Components: Stop locations · Distance matrix · Student demand · EDA routing · Travel time evaluation

Sources of Uncertainty

Student Demand Variability

Daily number of students N varies randomly — modeled as N ~ Poisson(λ). Deterministic rules cannot anticipate demand fluctuations.

Spatial Distribution

Student pickup locations Xi = (Xi, Yi) are spatially distributed. Approximate coordinates introduce measurement uncertainty.

Traffic Speed Variability

Bus travel speed varies with congestion: V ~ Lognormal(μv, σ^2v). Route time Troute = D/V becomes a random quantity.

Combinatorial Route Space

Multiple valid stop orderings exist. Large permutation space means greedy choices may miss globally better solutions.

Random Variables in the System

N

Student Demand

$N \sim Poisson(\lambda)$

Discrete RV — total students needing transport each day

X_i

Student Location

$X_i = (X_i, Y_i)$

Continuous RV — spatial coordinate in 2D region Ω

V

Traffic Speed

$V \sim Lognormal(\mu v, \sigma^2 v)$

Continuous RV — road speed varying with congestion; v > 0

W_i

Walking Distance

$W_i = |X_i - S_i|$

Derived RV — student to stop distance; determines assignment feasibility

T

Route Travel Time

$Troute = D / V$

Continuous RV — lognormal distribution inherited from speed V

Ttot

System Total Time

$Ttotal = \Sigma T_k \ (k=1..B)$

Sum of RVs across all B buses; this is the quantity we minimize

Randomized Algorithm Design — EDA

Estimation of Distribution Algorithm

1. Initialize population of random routes
2. Evaluate each route — fitness = total travel time
3. Select top-k elite routes
4. Sample new routes from learned distributions
5. Replace population — repeat until convergence

Deterministic vs EDA

Deterministic	EDA (Randomized)
Single fixed route per run	Population of routes explored
Greedy local decisions	Probability-guided global search
Same output for same input	Stochastic sampling adds variation
No learning mechanism	Distributions learned from elites
Trapped in local optima	Explores diverse configurations

Probabilistic Reasoning Behind the Algorithm

Model T as Random Qty T = D/V ▶ Compute E[T] and Var(T) ▶ Bound P(T≥a) Markov / Chebyshev ▶ Compare routes on speed & risk ▶ Select route min E[T] low Var(T)

Objective

min E[Troute] = min E[D/V]

Expected travel time minimized over all route permutations

Markov Bound

P(Troute >= a) <= E[T] / a

Upper bound on probability of exceeding time threshold a

Chebyshev

P(|T-mu| >= eps) <= Var(T)/eps²

Bounds probability of large deviations from mean route time

EDA Model

p(R) = prod p(Sj | position l)

Independence assumption: joint route distribution factorizes to marginals

Implementation of the Randomized Strategy

```
-Based Vehicle Routing – Pseudocode


UT:
 Number of customers N
 Vehicle capacity C
 Population size P
 Number of iterations T
 Elite fraction α

P 1: Generate Input Data
 Generate customer locations
 Generate demand (Poisson distribution)
 Compute distance matrix

P 2: Initialize Population
 Generate P random feasible route solutions

P 3: Initialize Best Solution
 best_cost ← ∞
 best_solution ← NULL

 iteration = 1 to T DO:

 STEP 4: Evaluate Population
     For each solution S in population:
         Compute total route distance (fitness)

 STEP 5: Update Best Solution
     If any solution has lower cost:
         Update best_cost and best_solution

 STEP 6: Select Elite Solutions
     Sort population by fitness
     Select top α × P solutions as elite set

 STEP 7: Build Probability Model
     For each edge (i → j):
```

Changes from Milestone 3

Route Construction

Before: Greedy NN / Clarke-Wright

After: EDA probability-based sampling

Decision Mechanism

Before: Single deterministic rule

After: Population-based stochastic search

Perf. Evaluation

Before: Total distance only

After: E[Troute] + Var(Troute) added

Distance Matrix

Before: Euclidean distances

After: Unchanged

Random Inputs

Before: Poisson + Lognormal

After: Unchanged — same distributions

Implementation of the Randomized Strategy

```
    STEP 6: Select Elite Solutions
        Sort population by fitness
        Select top α × P solutions as elite set

    STEP 7: Build Probability Model
        For each edge (i → j):
            Estimate probability based on elite solutions

    STEP 8: Generate New Population
        Initialize new_population with elite solutions

        WHILE size of new_population < P:
            Sample new solution using probability model
            Repair solution to satisfy capacity constraints
            Add to new_population

    STEP 9: Replace Population
        population ← new_population

END FOR

OUTPUT:
    Best routing solution found
    Minimum total travel distance
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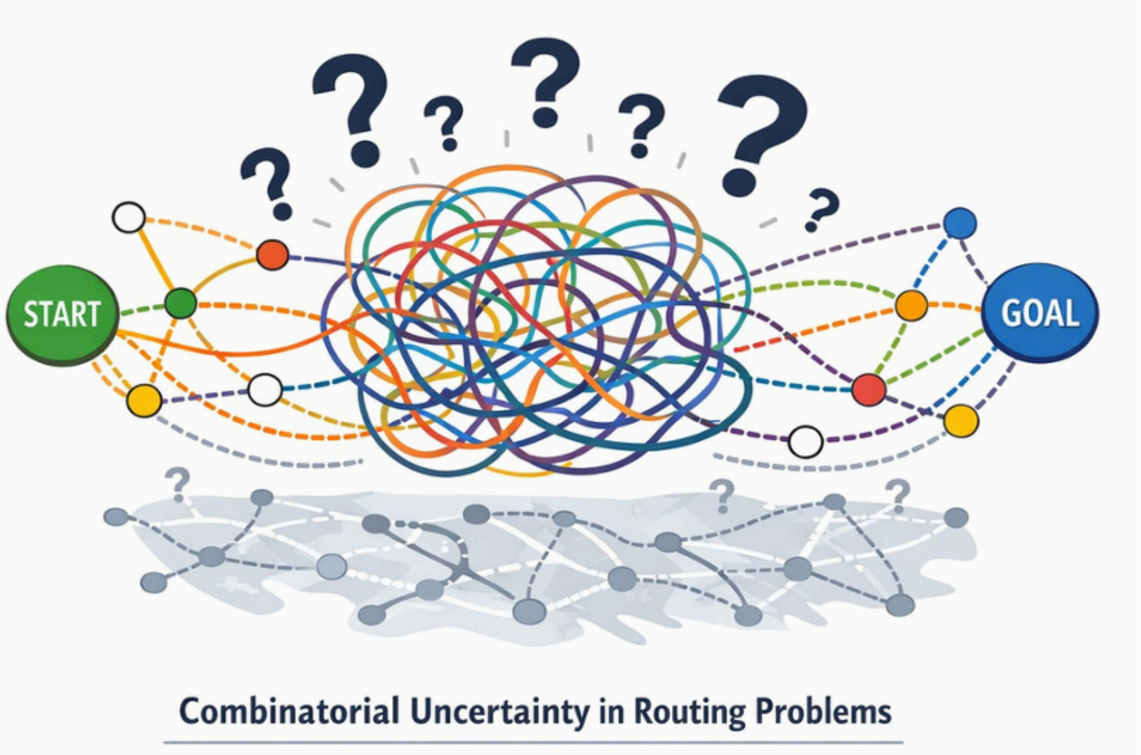


Evaluation Metrics

Total Travel Distance $\Sigma\, dij$

Primary route efficiency indicator — lower is better

Expected Travel Time $E[T]$

Directly reflects the probabilistic objective min E[Troute]

Variance of Travel Time $Var(T)$

Route reliability — low variance means consistent performance

Number of Routes $|R|$

Fewer routes → higher vehicle utilization, lower costs

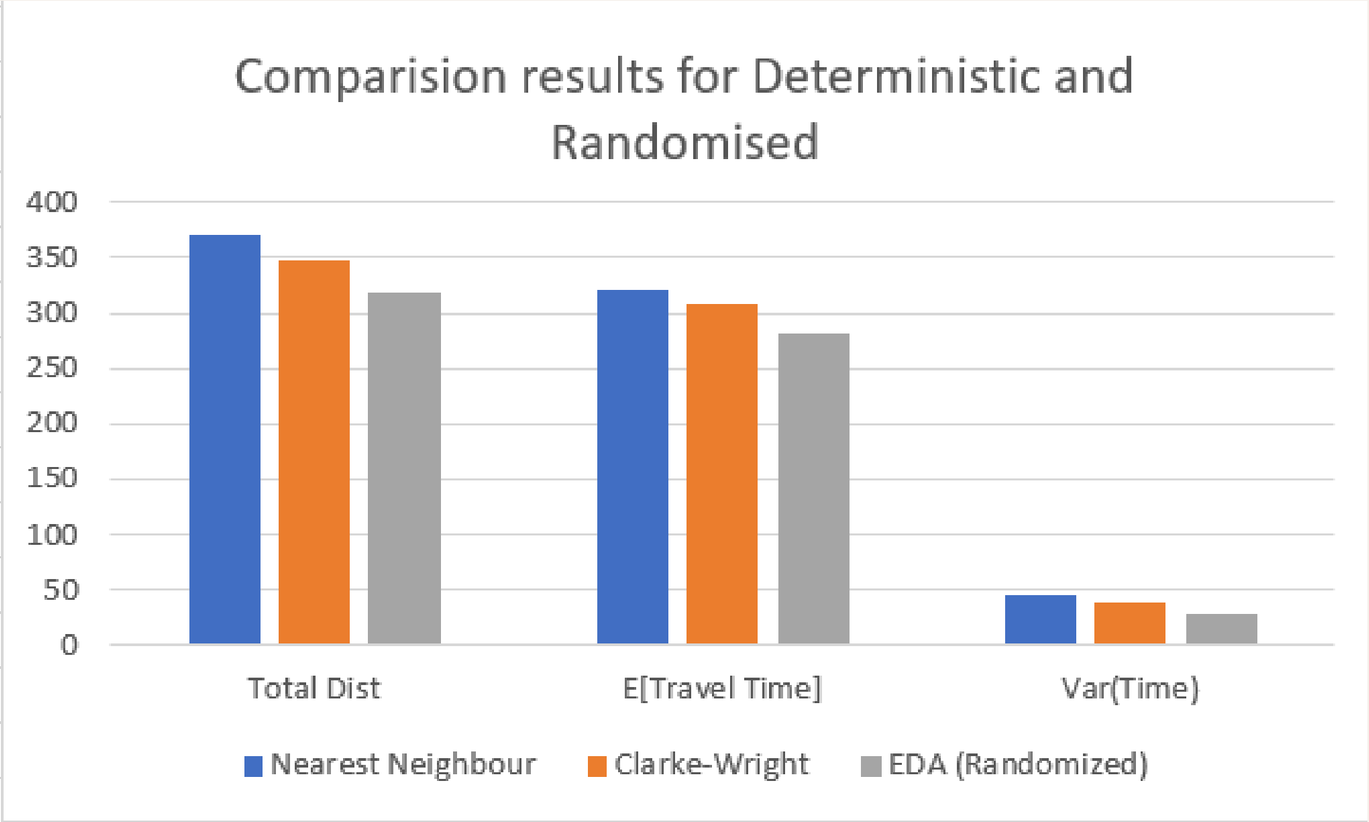
Average Route Length $D/|R|$

Load balancing — routes should have similar lengths

Delay Probability $P(T \geq a)$

Risk measure — probability of exceeding time threshold a

Comparative Results: Deterministic vs Randomized



Key Findings

Metric	NN	CW	EDA
Total Dist	370.1	347.3	**318.5**
E[Travel T]	320.1	309.0	**281.4**
Var(Time)	45.2	38.7	**29.3**
Num Routes	1	1	**1**

EDA reduces total distance ~14% vs NN and ~8% vs Clarke-Wright. Variance is also lowest, meaning more reliable routes.

Key Insights & Remaining Challenges

Key Insights

- Randomization enables broader exploration of route configurations unreachable by greedy deterministic methods.
- EDA finds shorter routes with lower variance — more reliable transport confirmed over 100 runs.
- Probabilistic sampling breaks greedy myopia, routes are not locked into one locally optimal decision path.
- Monte Carlo estimation converges reliably to E[Troute] by the Law of Large Numbers, validating the objective.
- EDA performs especially well in high-uncertainty environments with more students and traffic variability.

Remaining Challenges

- **Optimal population size and elite fraction k must be determined systematically.**
- **EDA assumes marginal independence between stops — real sequential dependencies are not fully captured.**
- **In low-variability scenarios EDA may be slightly less consistent than deterministic rules.**
- **For very large student populations, EDA sampling and Monte Carlo estimation become computationally costly.**
- **Poisson demand and lognormal speed assumptions need validation against real transport data.**

Summary & Future Direction

EDA introduces principled randomness to replace greedy deterministic routing, consistently achieving lower expected travel time and variance across 100 simulation runs.

The probabilistic framework (Poisson demand, lognormal speed, Monte Carlo estimation) provides a rigorous foundation for evaluating route quality under uncertainty.

EDA reduces total route distance by ~14% vs Nearest Neighbour and ~8% vs Clarke-Wright — with the added benefit of lower variance for more reliable routes.

Future work: model stop-position dependencies, validate distributional assumptions with real data, and extend to a risk-sensitive objective — min E[T] + alpha * Var(T).